

# Week 1 - Workshop

## C# Introduction and Setup

# Visual Studio Installation

1. Visual Studio Installation
2. Rider Installation
3. Creating a Console App (Visual Studio)
4. Creating a Console App (Rider)
5. Lab Task

# Visual Studio Installation

## System Requirements for Visual Studio:

- Windows OS
- Minimum of 4GB RAM
- 20 - 25 GB of storage space
- x64 processor
- [Check more requirements here.](#)

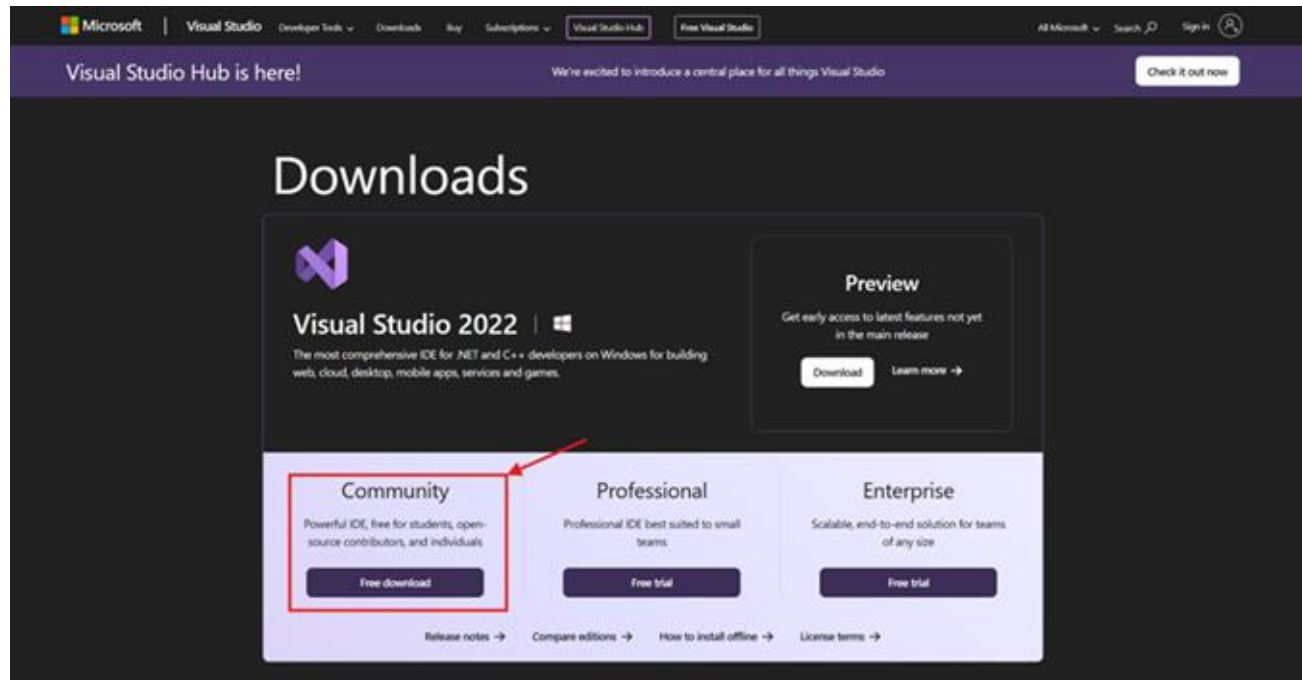
1. Visual Studio Installation
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# Visual Studio Installation

## Visual Studio installation steps:

- Click [here](#) to be redirected to installation page or copy paste following URL:

**Download link:** <http://visualstudio.microsoft.com/downloads/>



## 1. Visual Studio Installation

## 2. Rider Installation

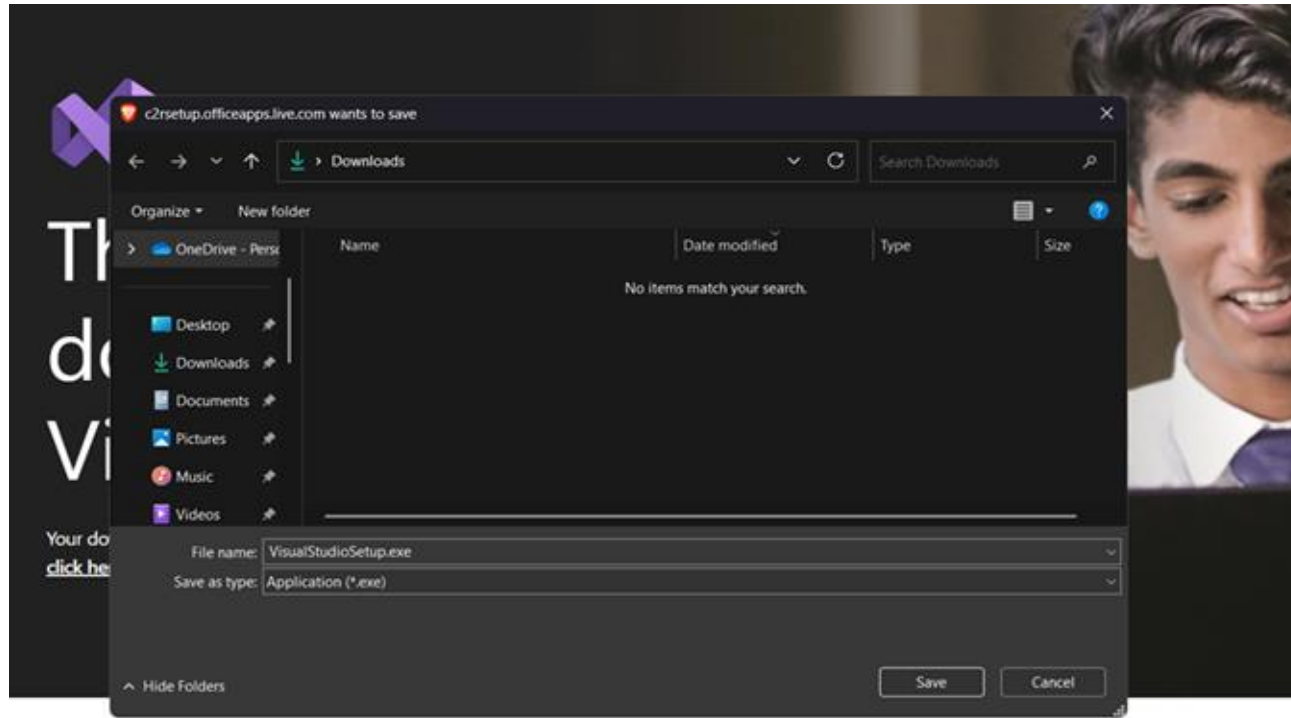
## 3. Creating a Console App (Visual Studio)

## 4. Creating a Console App (Rider)

## 5. Lab Task

# Visual Studio Installation

- Save the **exe** file.

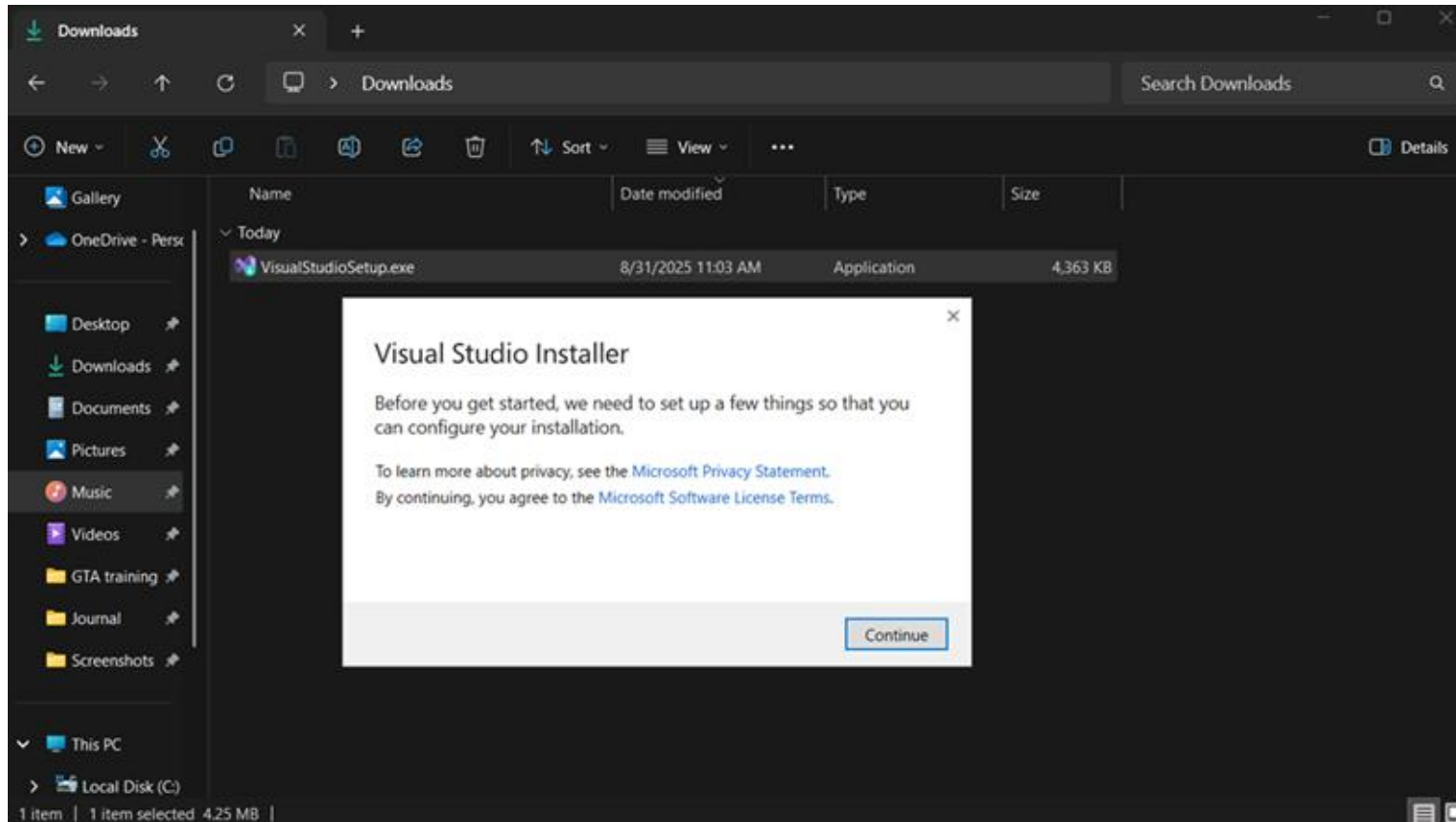


New to Visual Studio?

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# Visual Studio Installation

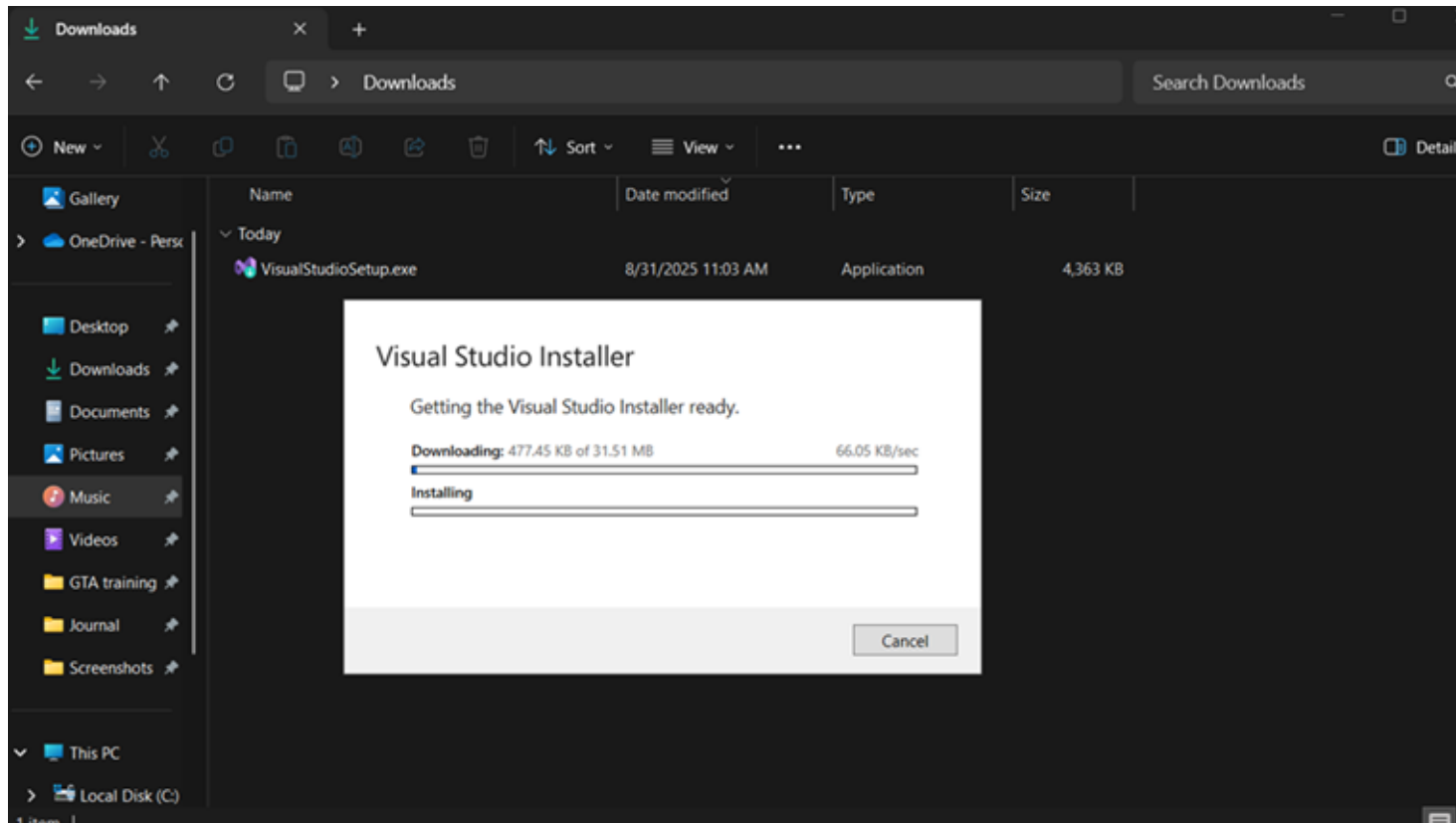
- Open the exe file and click on continue.



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# Visual Studio Installation

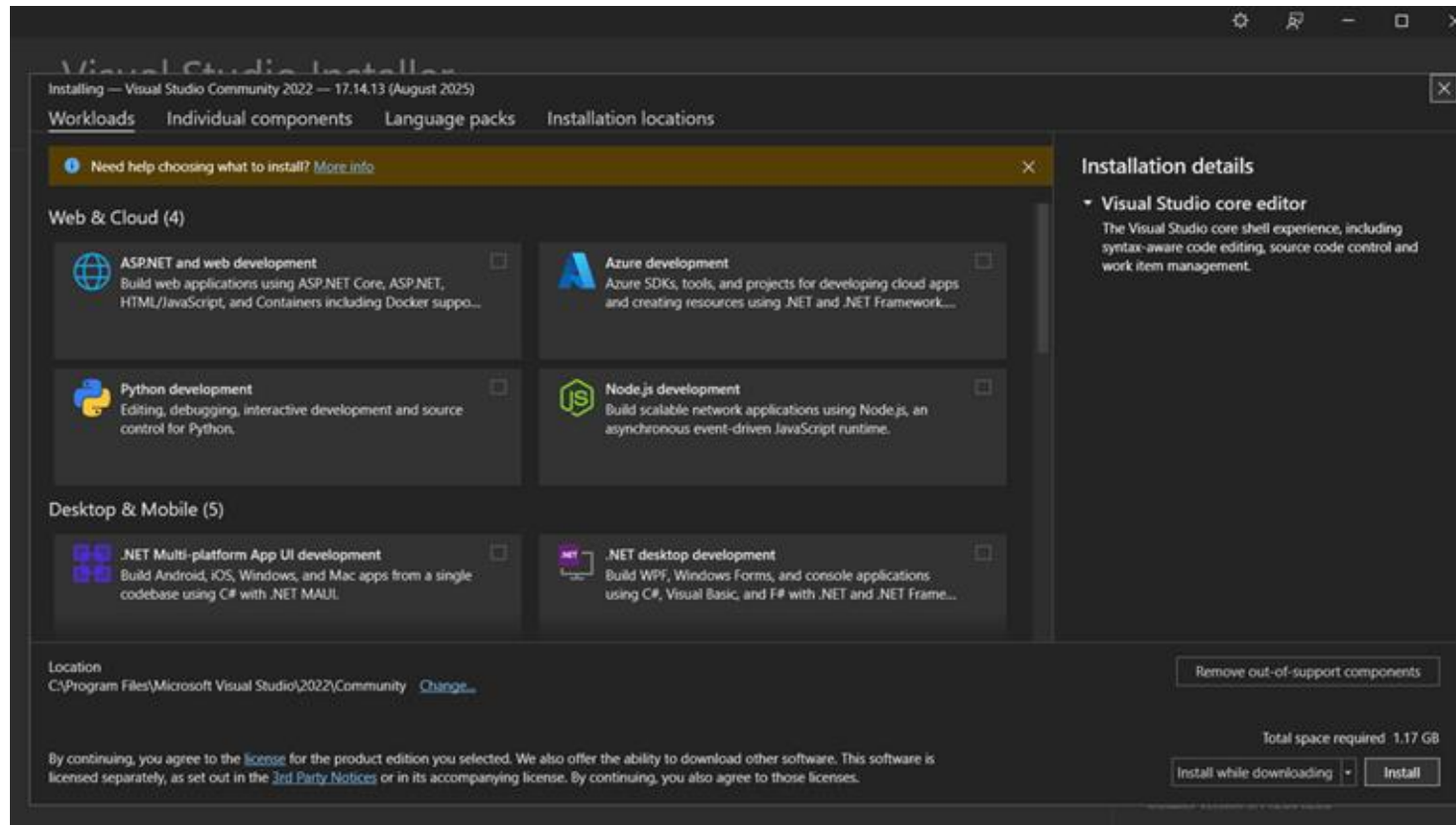
- Wait for the Visual Studio installer to be installed.



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# Visual Studio Installation

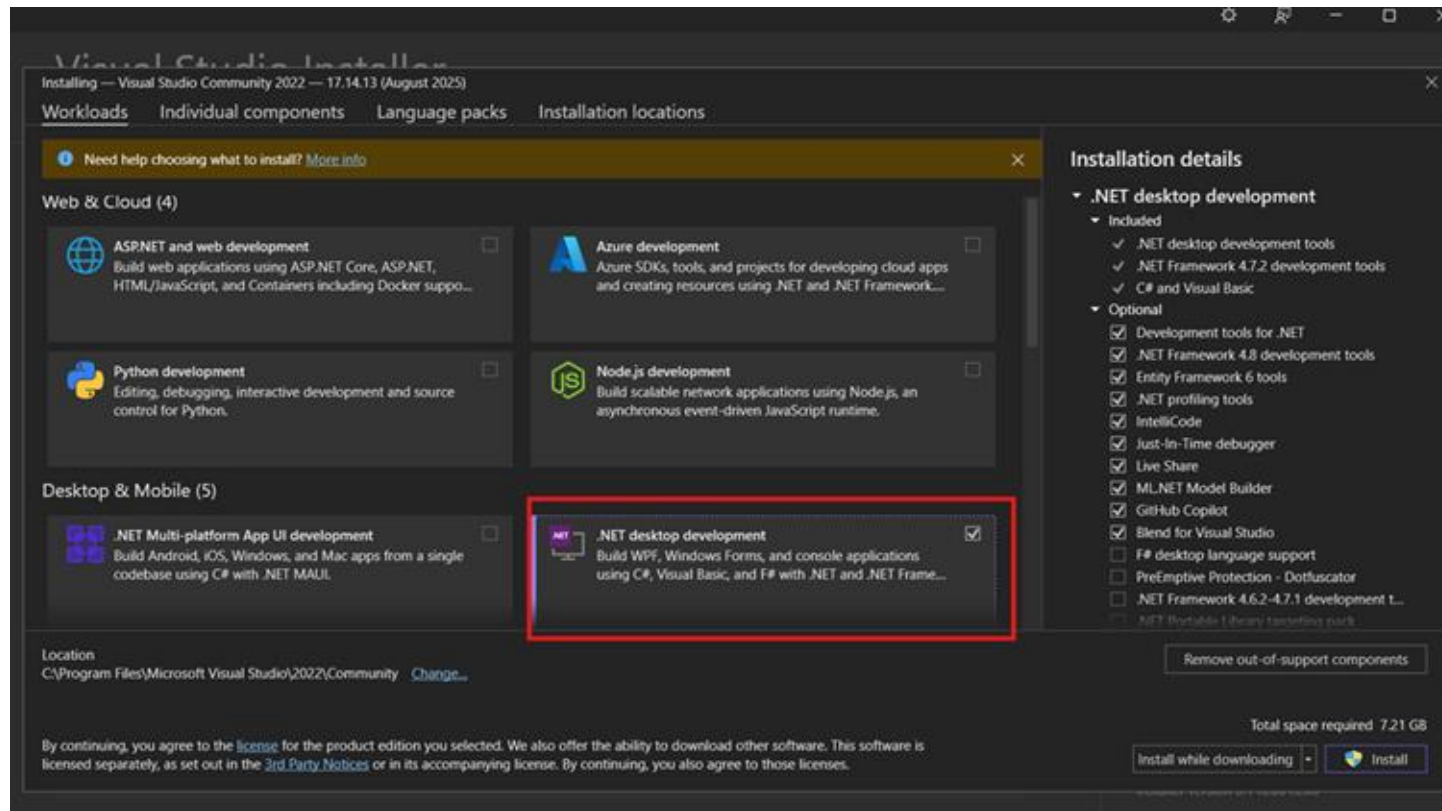
- Once the installer is installed, you will see a dialog box similar to below:



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# Visual Studio Installation

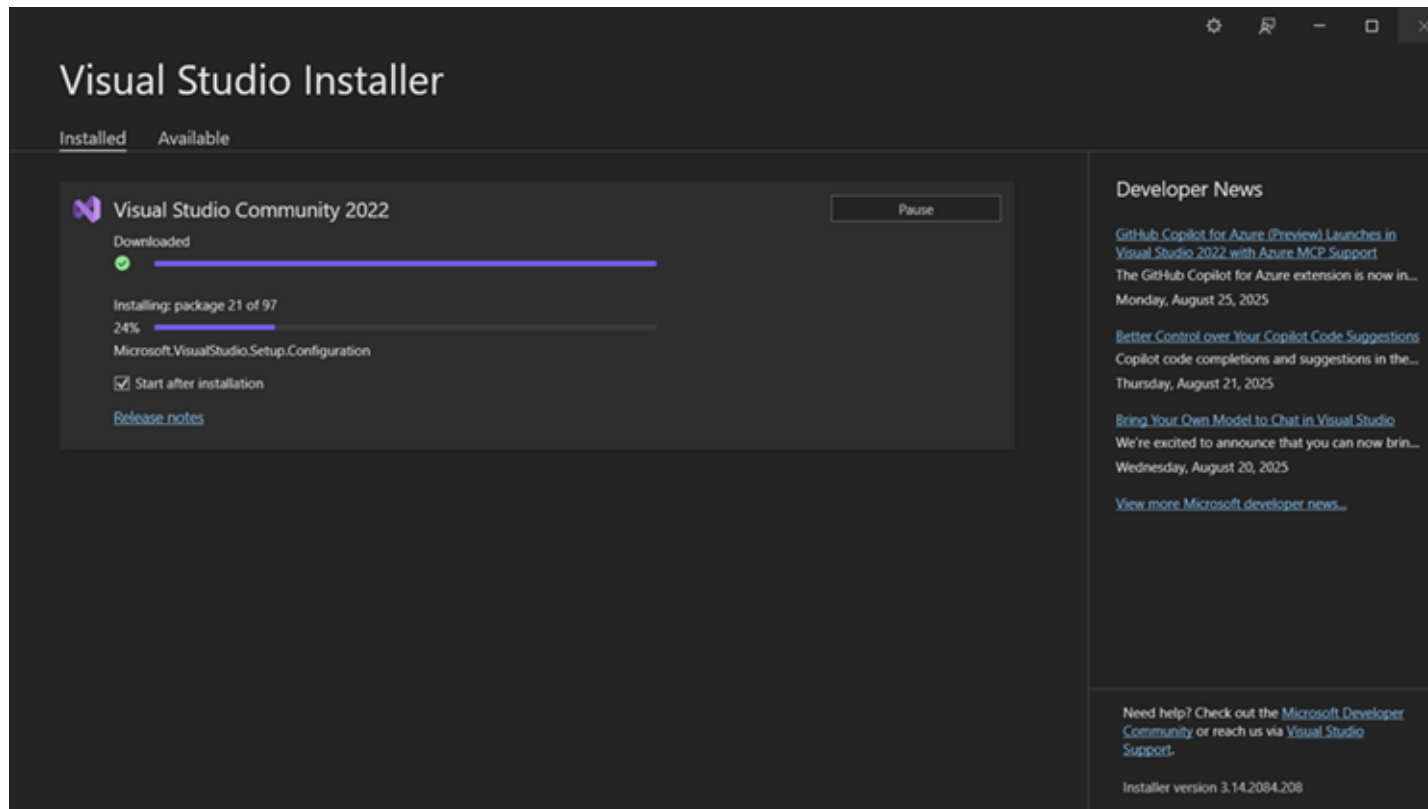
- For working with console-based applications, you would need to install the following package: **".NET desktop development"**. Then click on install



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# Visual Studio Installation

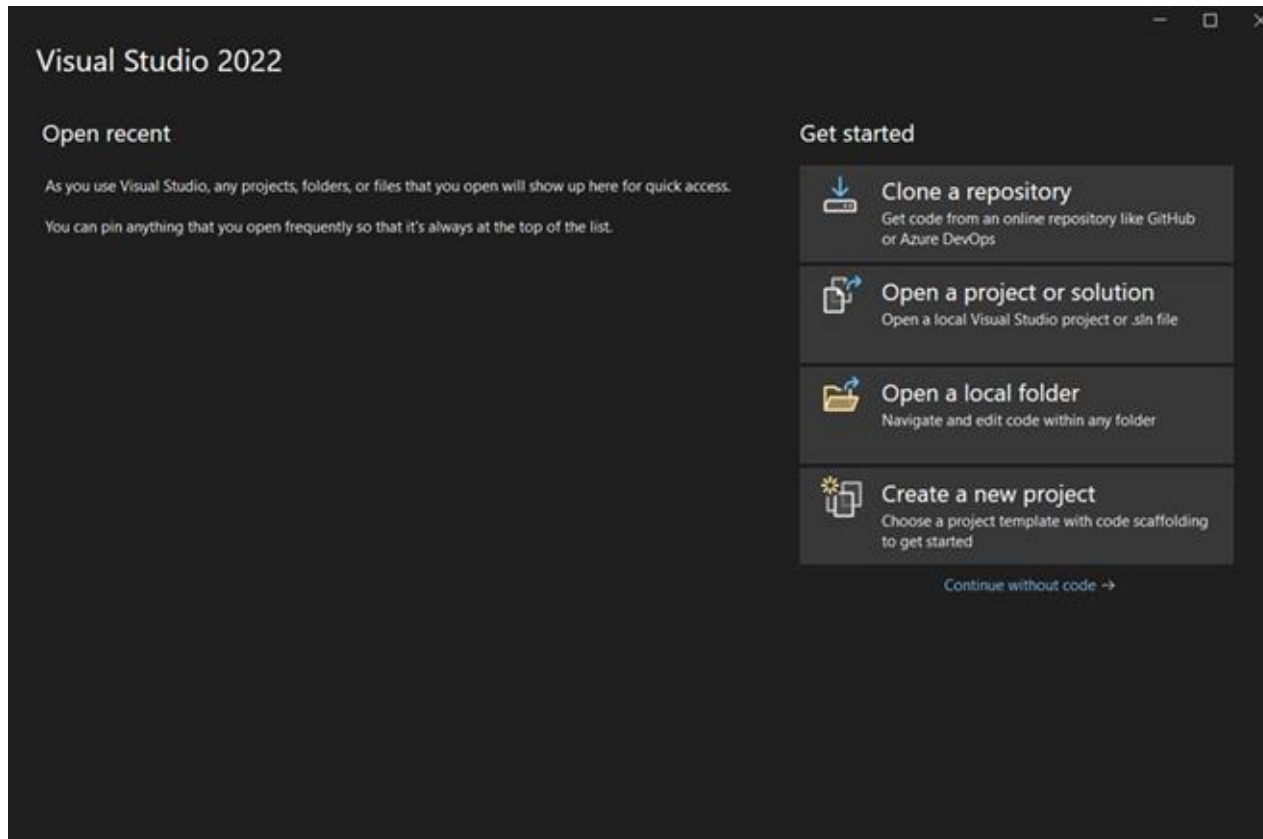
- Wait for the installation to be completed.



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# Visual Studio Installation

- Excellent, now you have taken first step into becoming a .NET developer.



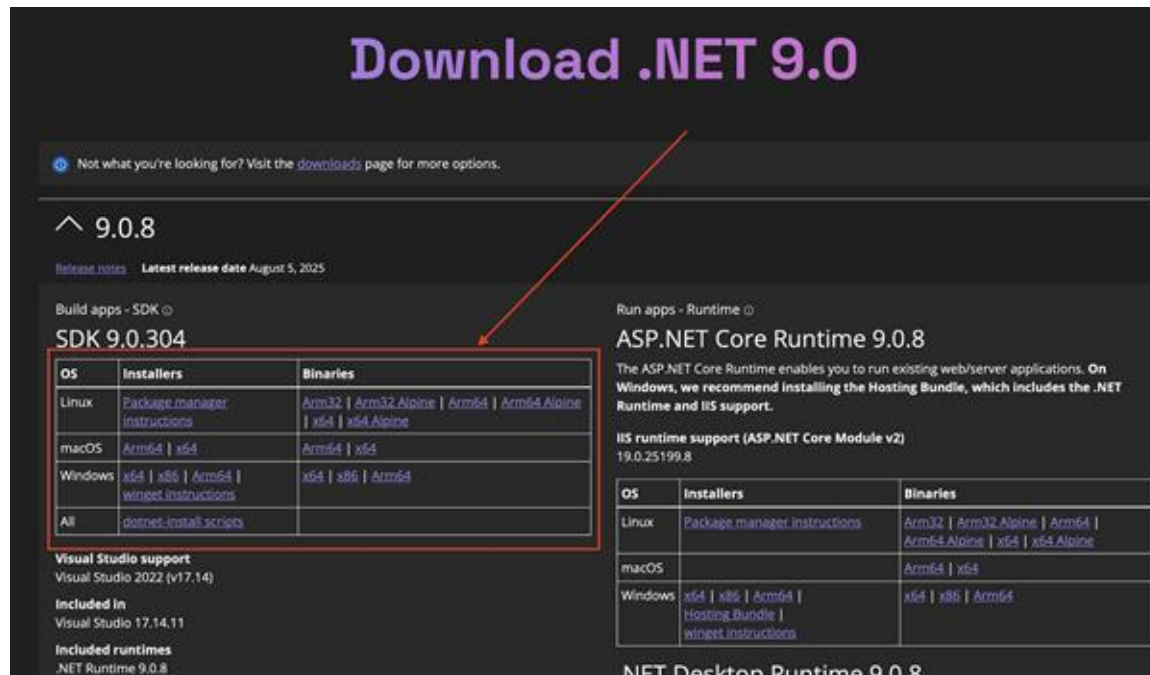
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# Rider Setup ( For Mac & Linux)

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# Rider Setup( For Mac & Linux)

- Go to <https://dotnet.microsoft.com/en-us/download/dotnet/9.0>
- Here navigate the the sdk section



The screenshot shows the 'Download .NET 9.0' page. A red arrow points to the 'SDK 9.0.304' section. Below it is a table with columns 'OS', 'Installers', and 'Binaries'. The table lists installation instructions for Linux, macOS, Windows, and All. To the right, there is a section for 'ASP.NET Core Runtime 9.0.8' and 'IIS runtime support (ASP.NET Core Module v2)'. At the bottom, there is a section for 'Visual Studio support'.

| OS      | Installers                              | Binaries                                        |
|---------|-----------------------------------------|-------------------------------------------------|
| Linux   | Package manager instructions            | Arm32   Arm32 Alpine   Arm64   x64   x64 Alpine |
| macOS   | Arm64   x64                             | Arm64   x64                                     |
| Windows | x64   x86   Arm64   winget instructions | x64   x86   Arm64                               |
| All     | dotnet-install scripts                  |                                                 |

**Visual Studio support**  
Visual Studio 2022 (v17.14)  
**Included in**  
Visual Studio 17.14.11  
**Included runtimes**  
.NET Runtime 9.0.8

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# Rider Setup( For Mac & Linux)

## macOS Installers

### • Arm64

- For Apple Silicon Macs (M1, M2, M3, etc.).
- Works on macOS 11 (Big Sur) and later.
- Optimized for performance on ARM-based Apple processors.

### • x64

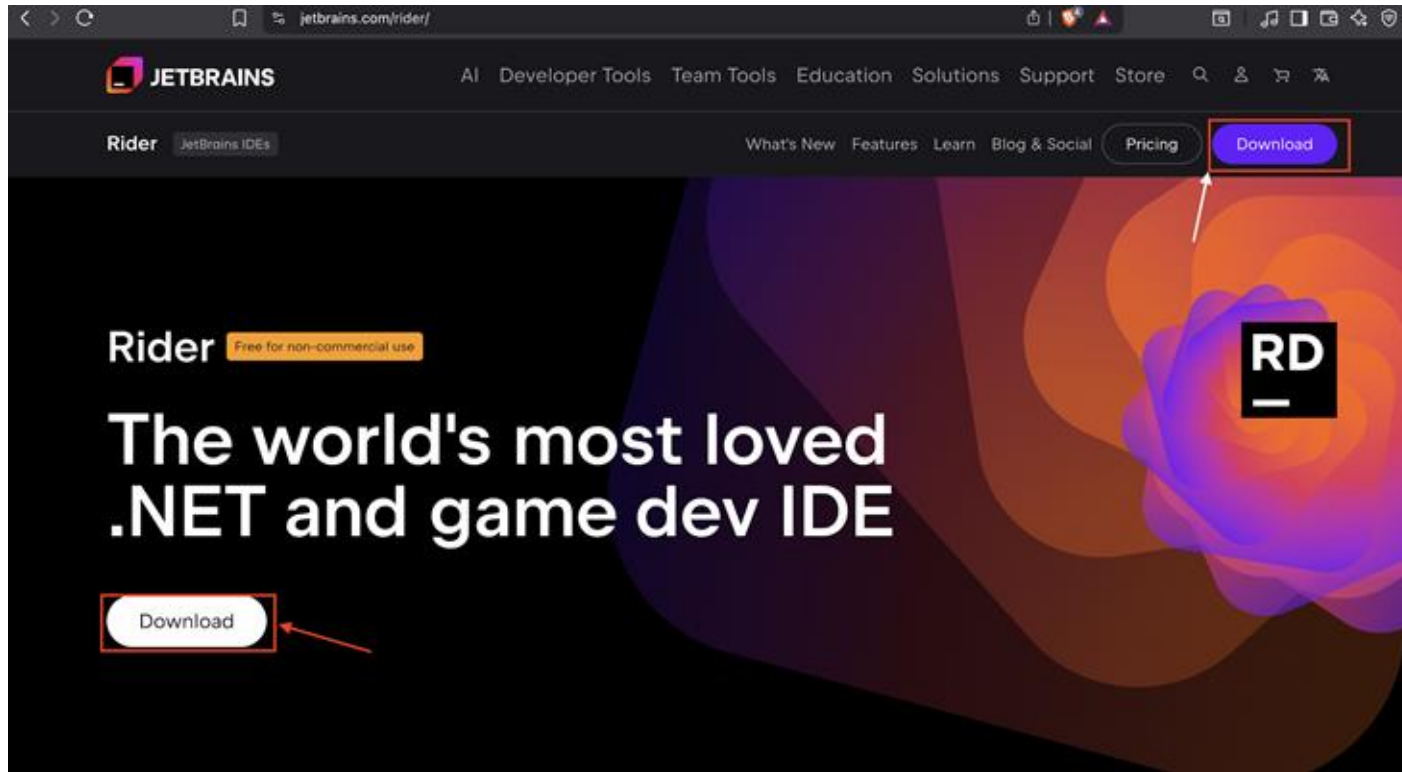
- For Intel-based Macs.
- Works on macOS 10.15 (Catalina) and later.
- Runs natively on Intel chips but can also run on Apple Silicon via Rosetta 2.

**NOTE: Before setting up rider, this step is a must.**

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# Rider Setup( For Mac & Linux)

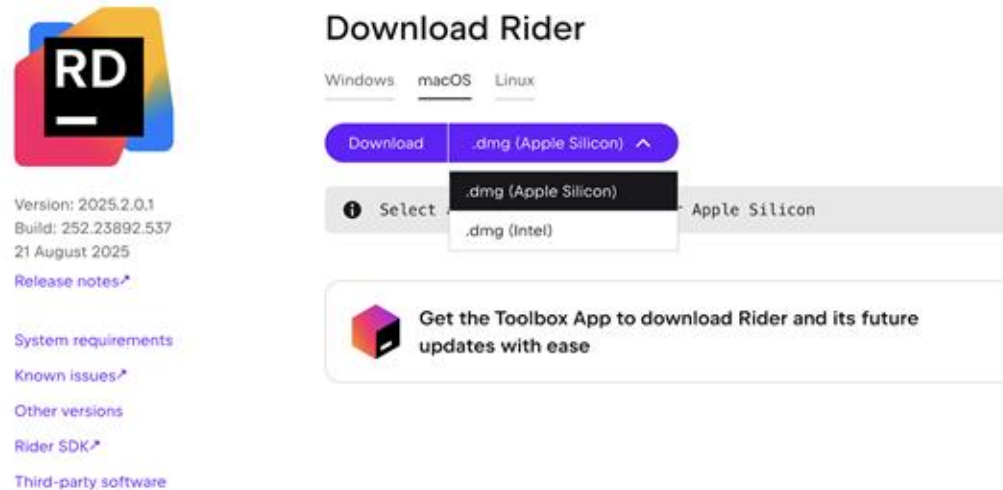
- Go to [jetbrains.com/rider/](https://jetbrains.com/rider/) and select any of the download option shown below



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# Rider Setup( For Mac & Linux)

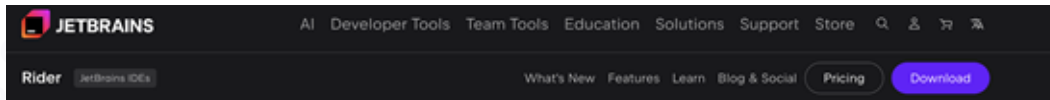
- Here we will be given two options to download:
  - Select the **.dmg (Apple Silicon)** version if your **Mac** has an **M1 or upper** version chip
  - Select the **.dmg (Intel)** version if your **Mac** uses an **Intel-based processor**.



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# Rider Setup( For Mac & Linux)

- You will soon be redirected to another page for an automatic download. If the download does not start automatically, please choose the manual download option.



## Thank you for downloading Rider!

Your download should start shortly. If it doesn't, please use the [direct link](#).

Download and verify the file [SHA-256 checksum](#).

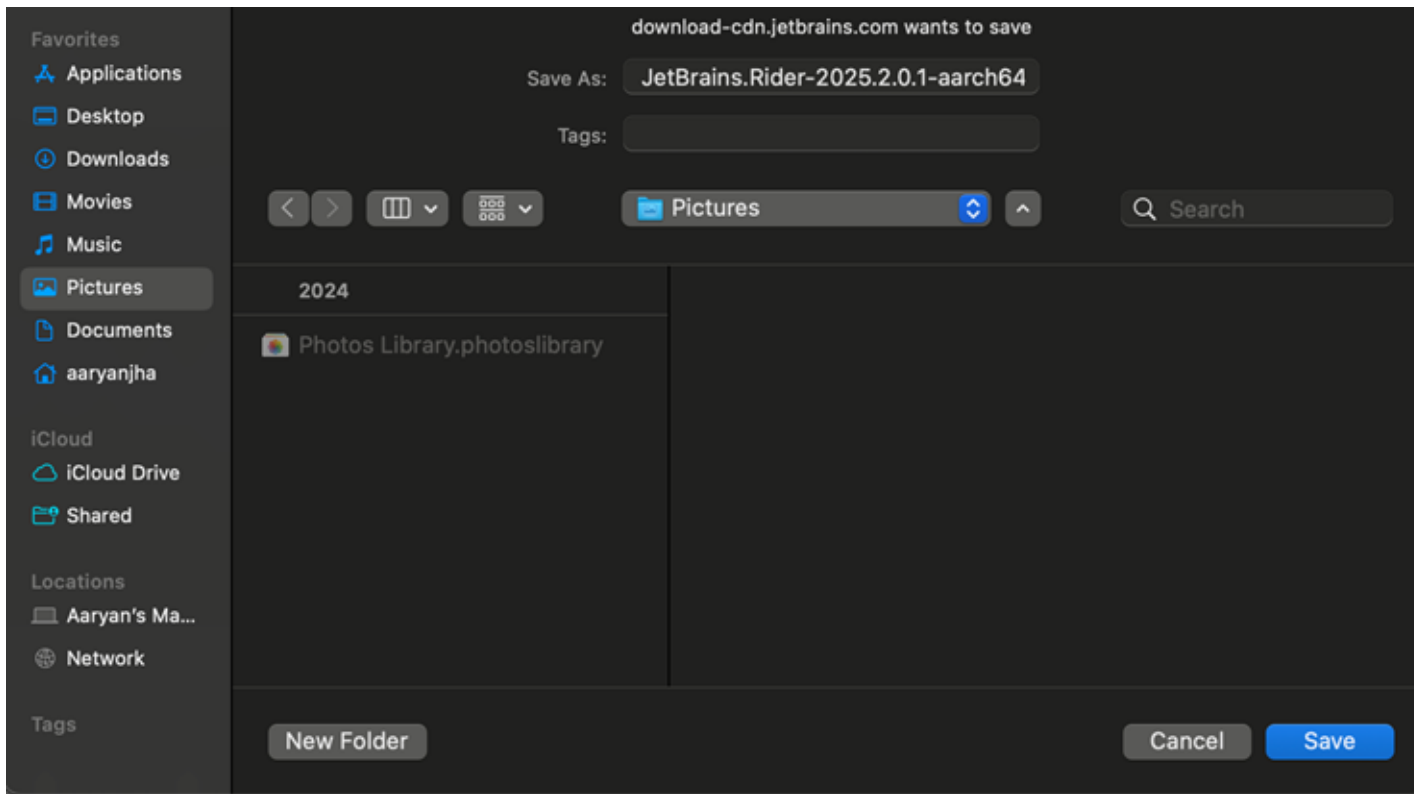
[Learn more about the digital signatures of JetBrains binaries.](#)

- + Can I use Rider for free?
- + Is the trial free?
- + I want to try out JetBrains products in my company, but I cannot use the internet from our corporate network (a secure environment is used, products are evaluated offline, etc.). How can I do this?
- + How can I make my trial more effective?

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# Rider Setup( For Mac & Linux)

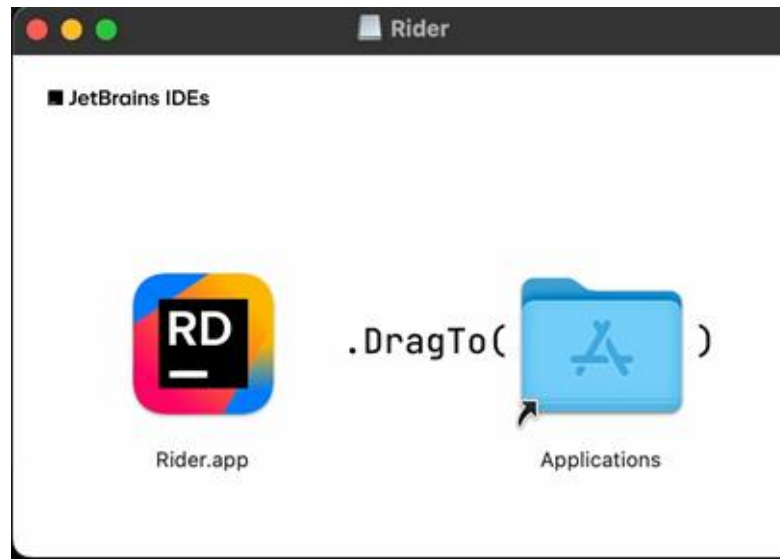
- Select the directory to **save** the setup file and hit save.



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# Rider Setup( For Mac & Linux)

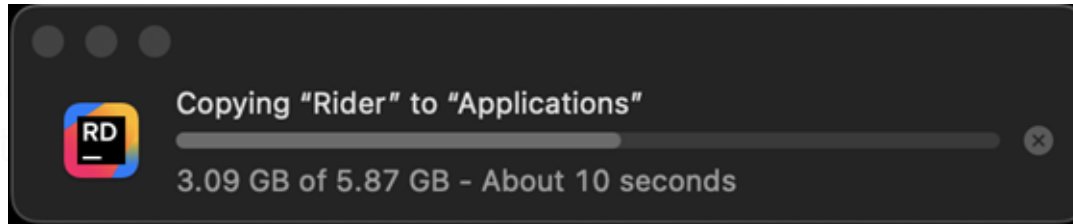
- Now open the setup installer after successful download then drag and drop the rider.app to applications folder.
- **Please take reference from the video below:**



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# Rider Setup( For Mac & Linux)

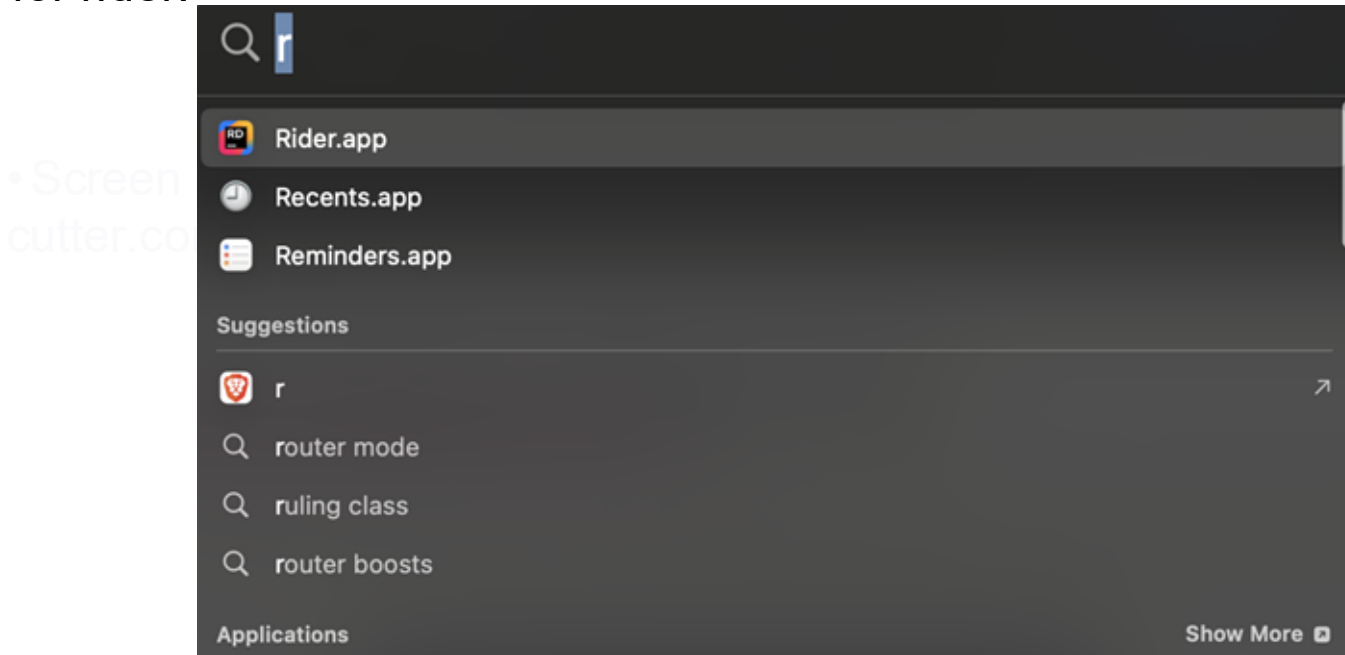
- After that, a pop up will appear regarding copying of application files



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# Rider Setup( For Mac & Linux)

- Finally, after successful copying press **Command + space** arrow button and search for rider.

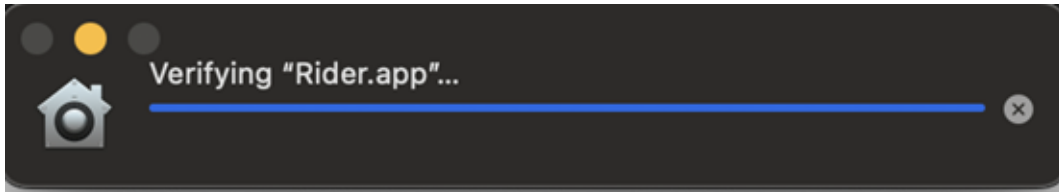


- Select **Rider.app** from the list of search results and hit **enter**.

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# Rider Setup( For Mac & Linux)

- During first launch, the application files are verified.



- After successful verification, a confirmation prompt might appear.

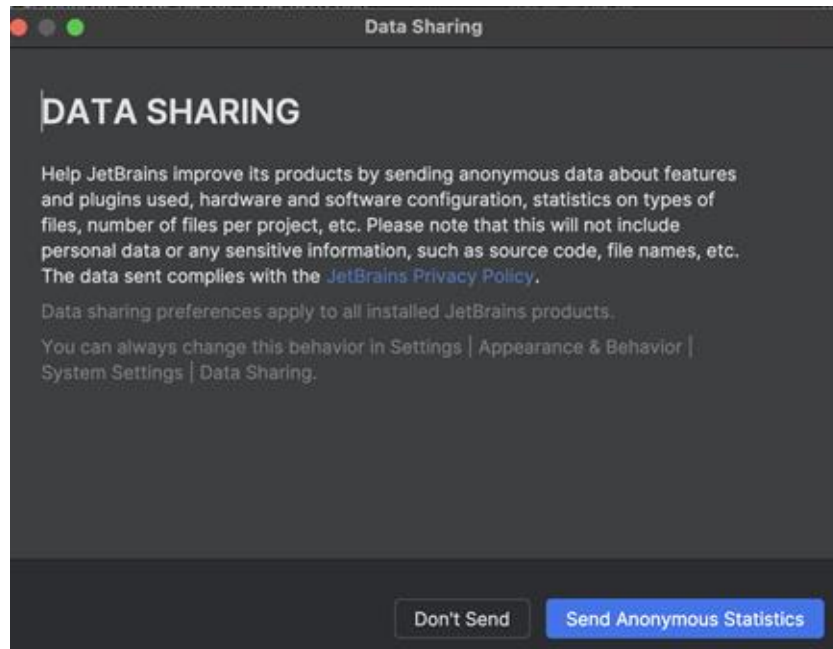


- Hit open.

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# Rider Setup( For Mac & Linux)

- After this another prompt will appear regarding data sharing

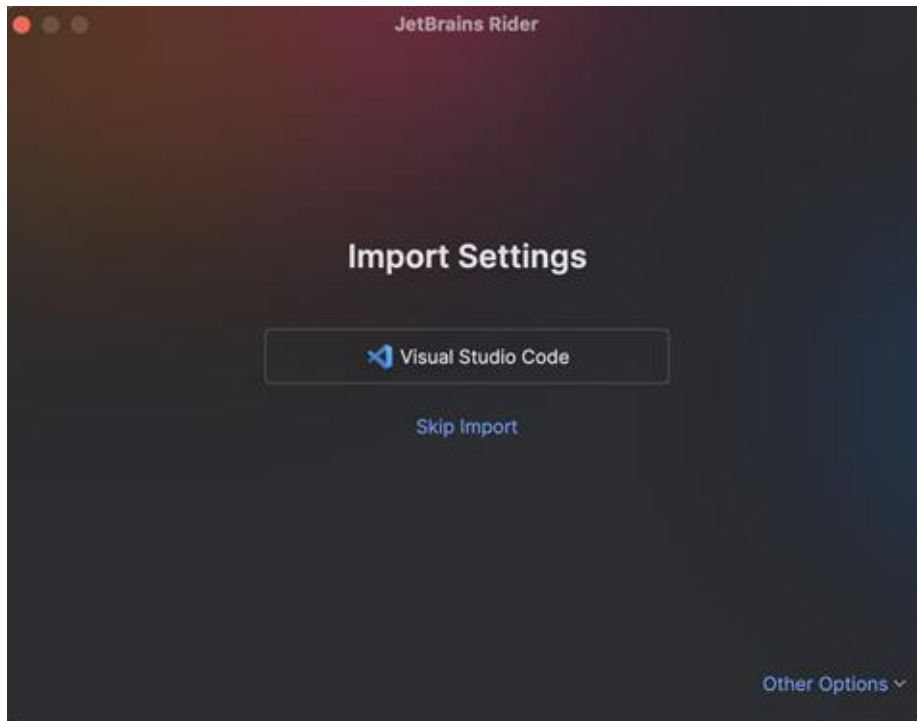


- It is recommended to select the option **Don't Send**

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# Rider Setup( For Mac & Linux)

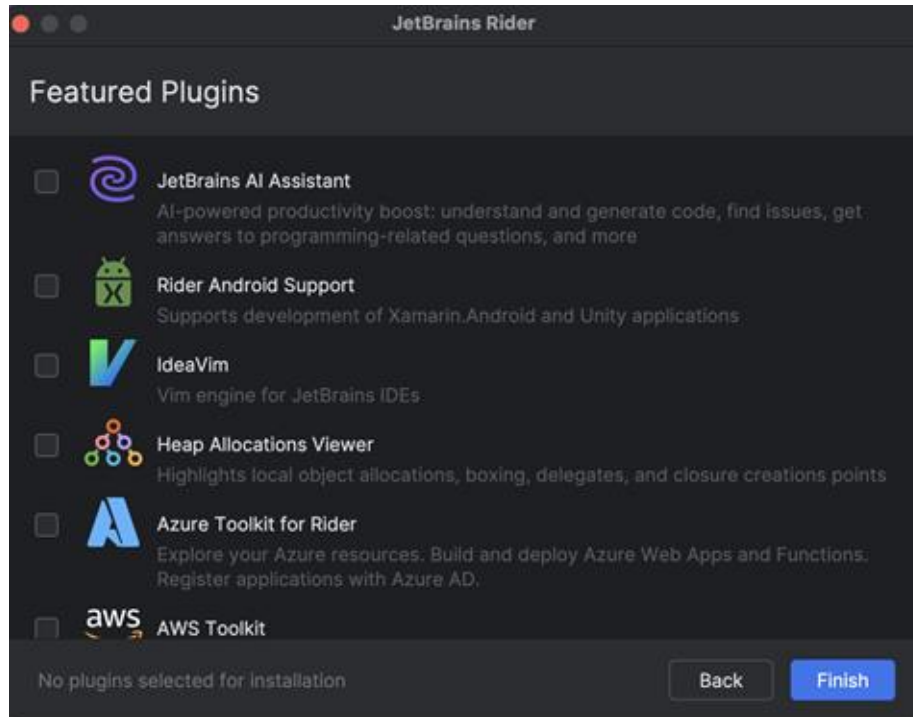
- Next, a prompt will appear asking if you'd like to import settings from another IDE. You can choose based on your preference, as this will copy keybindings and configurations from your previous IDE to this one.



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# Rider Setup( For Mac & Linux)

- Now, a final prompt will appear regarding the installation of extra dependencies.



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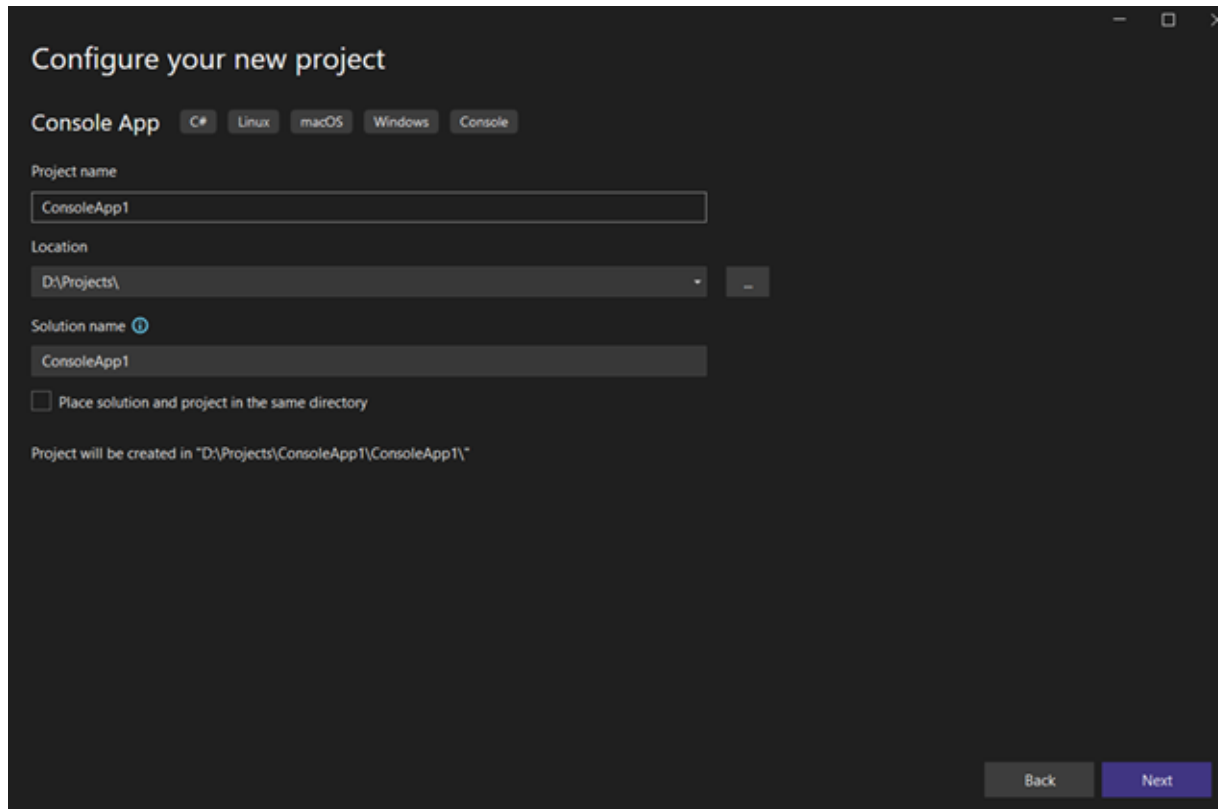
- As per **our scope** we don't require any of it. So, we will just hit finish without selecting any.

# Creating a .NET Console App

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# Creating a .NET Console App (Visual Studio)

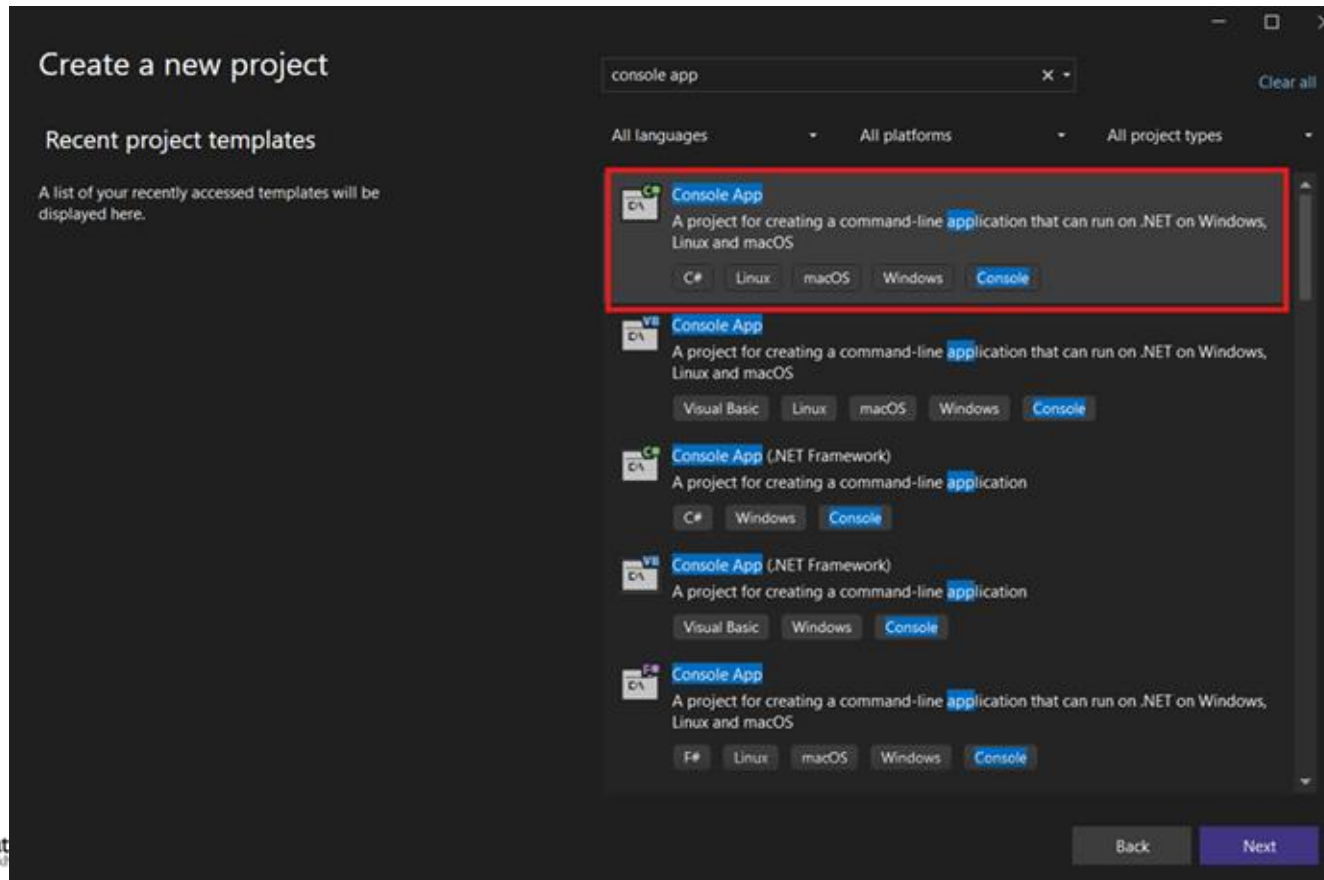
- Now enter the name of the project as you want, choose where you want to create the project then click on next.



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# Creating a .NET Console App (Visual Studio)

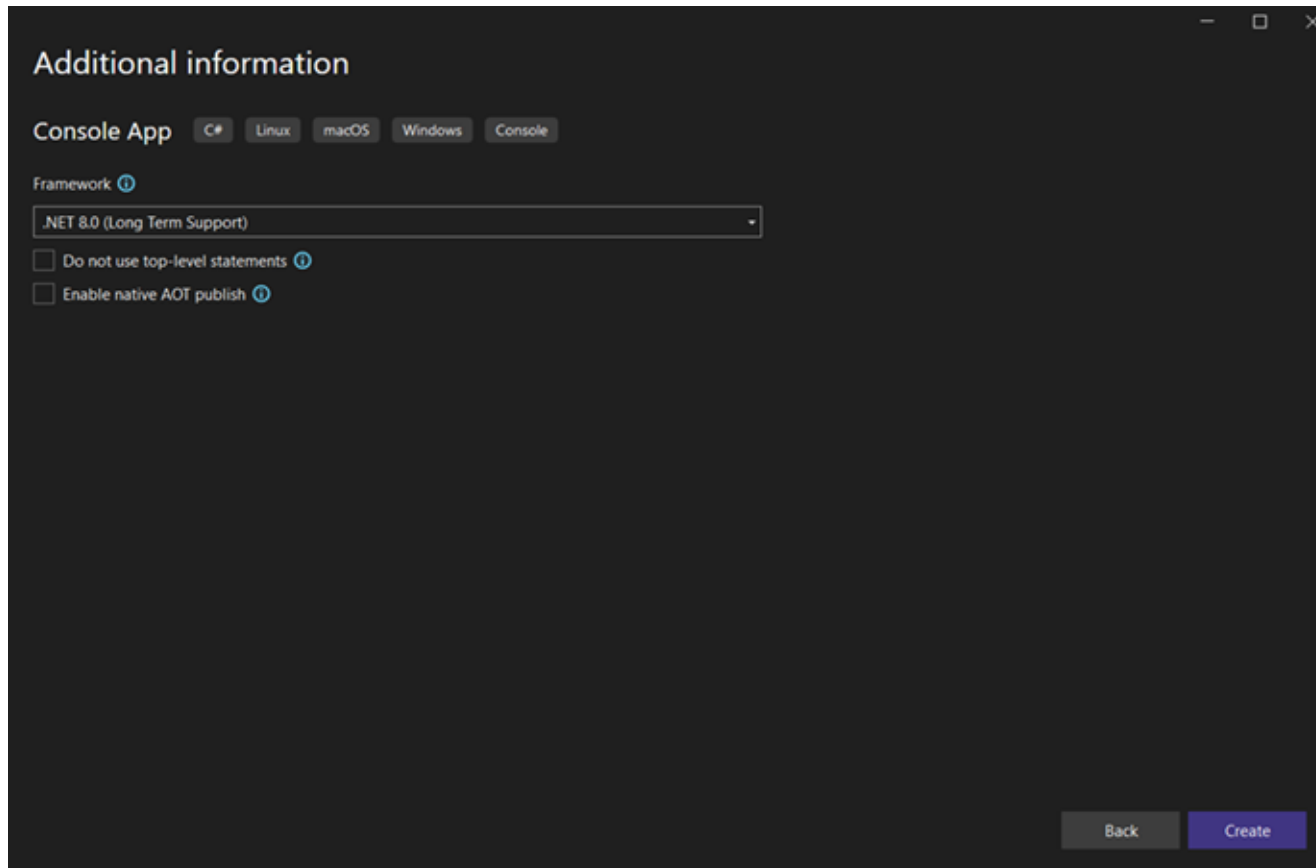
- Search for console app in the search bar and select the following highlighted template.



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# Creating a .NET Console App (Visual Studio)

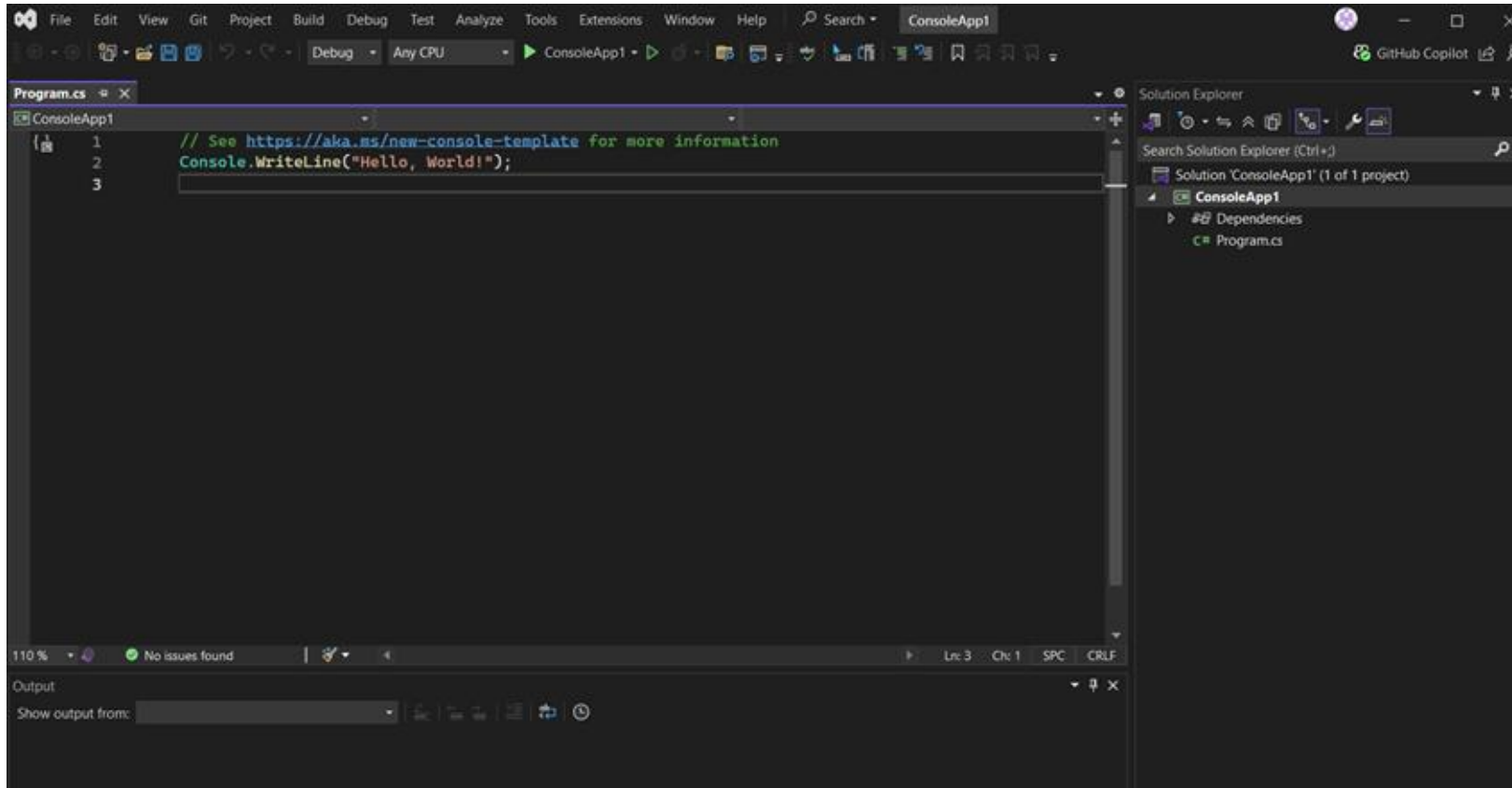
- Now click on **create**



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# Creating a .NET Console App (Visual Studio)

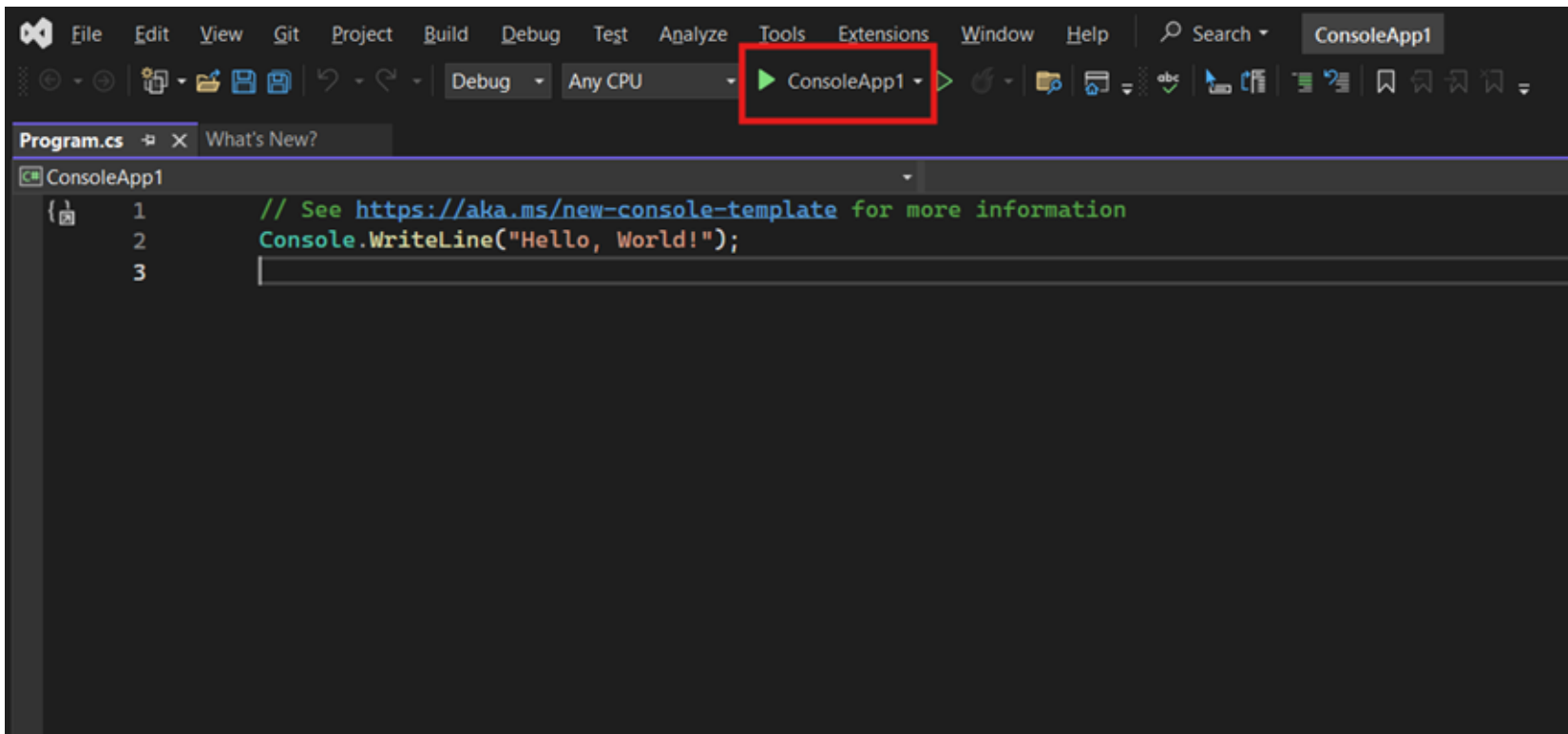
- Now you can start writing your code.



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# Creating a .NET Console App (Visual Studio)

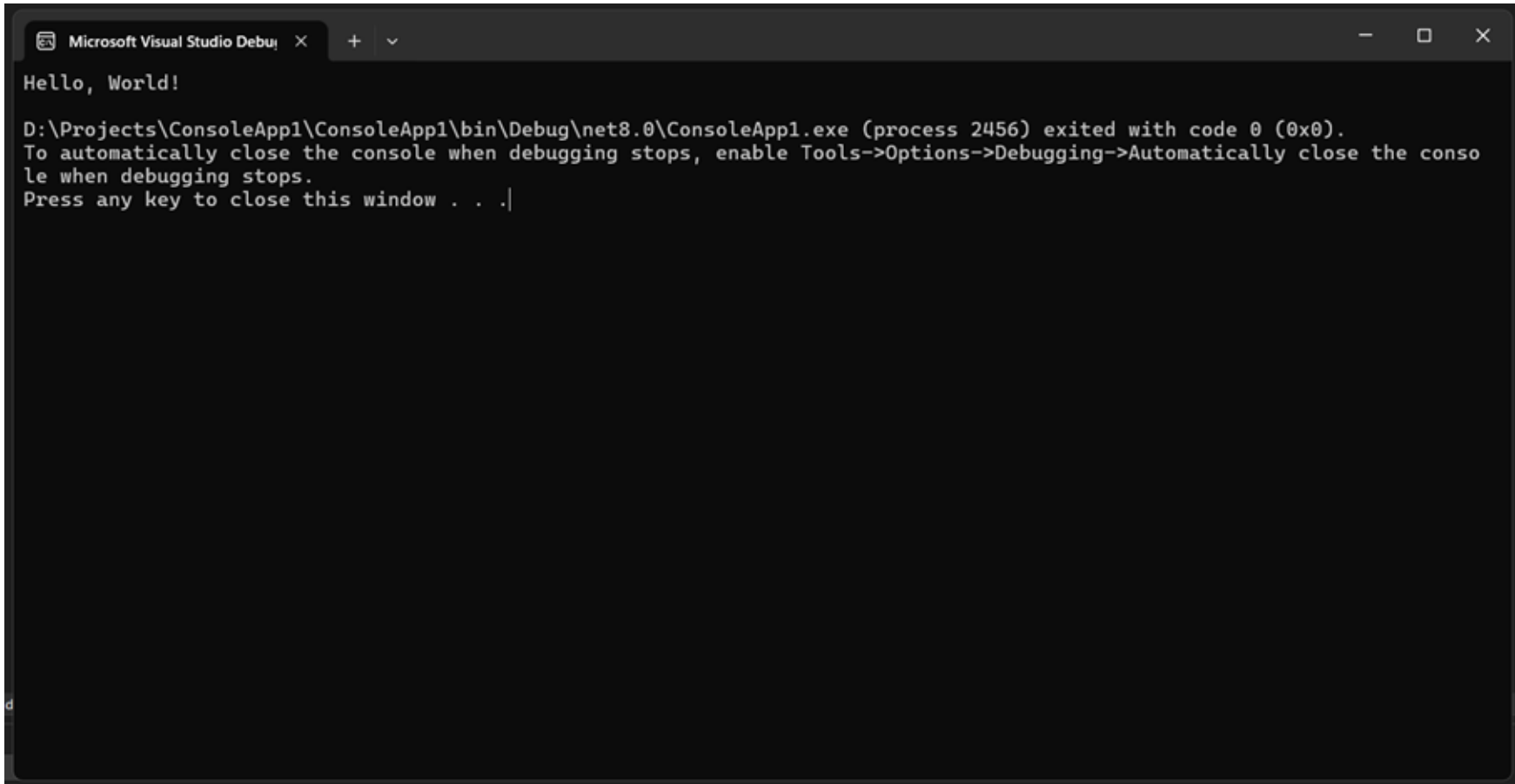
- For running your console app click on the following highlighted button



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# Creating a .NET Console App (Visual Studio)

- A new terminal will now open with the output



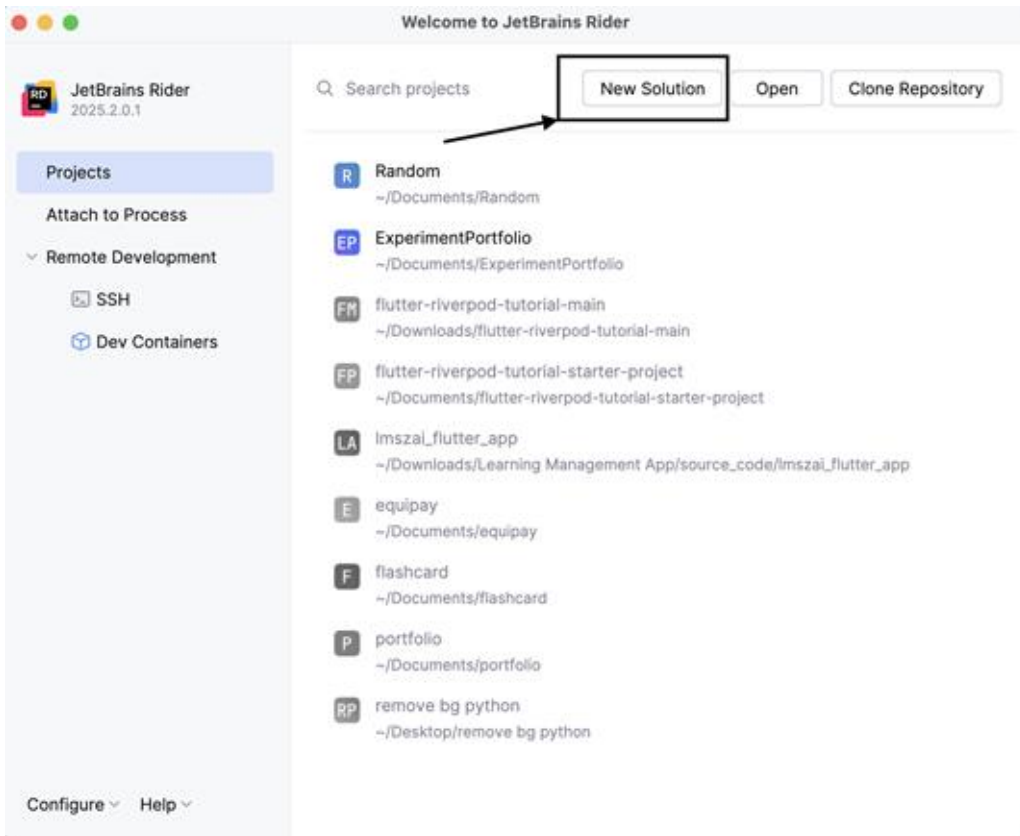
The screenshot shows a 'Microsoft Visual Studio Debug Console' window. The output text is as follows:

```
Hello, World!  
  
D:\Projects\ConsoleApp1\ConsoleApp1\bin\Debug\net8.0\ConsoleApp1.exe (process 2456) exited with code 0 (0x0).  
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.  
Press any key to close this window . . .|
```

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# Creating a Console App (Rider)

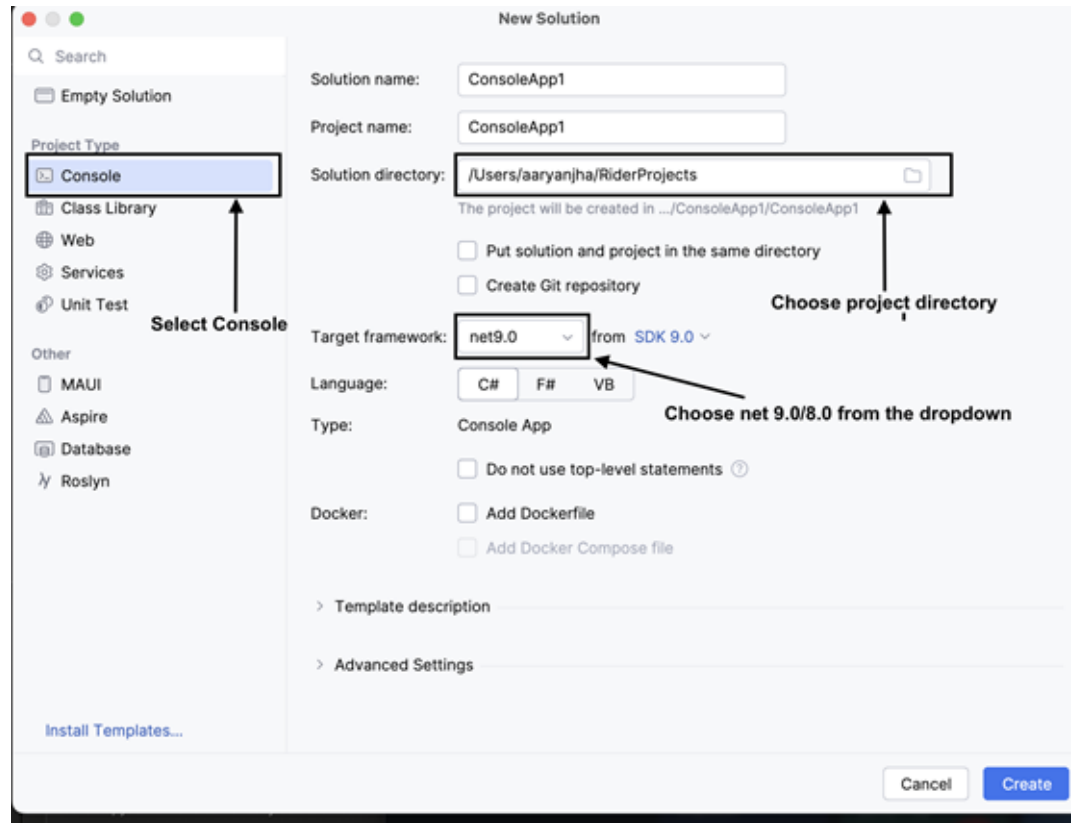
- Here, create a new solution.



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# Creating a Console App (Rider)

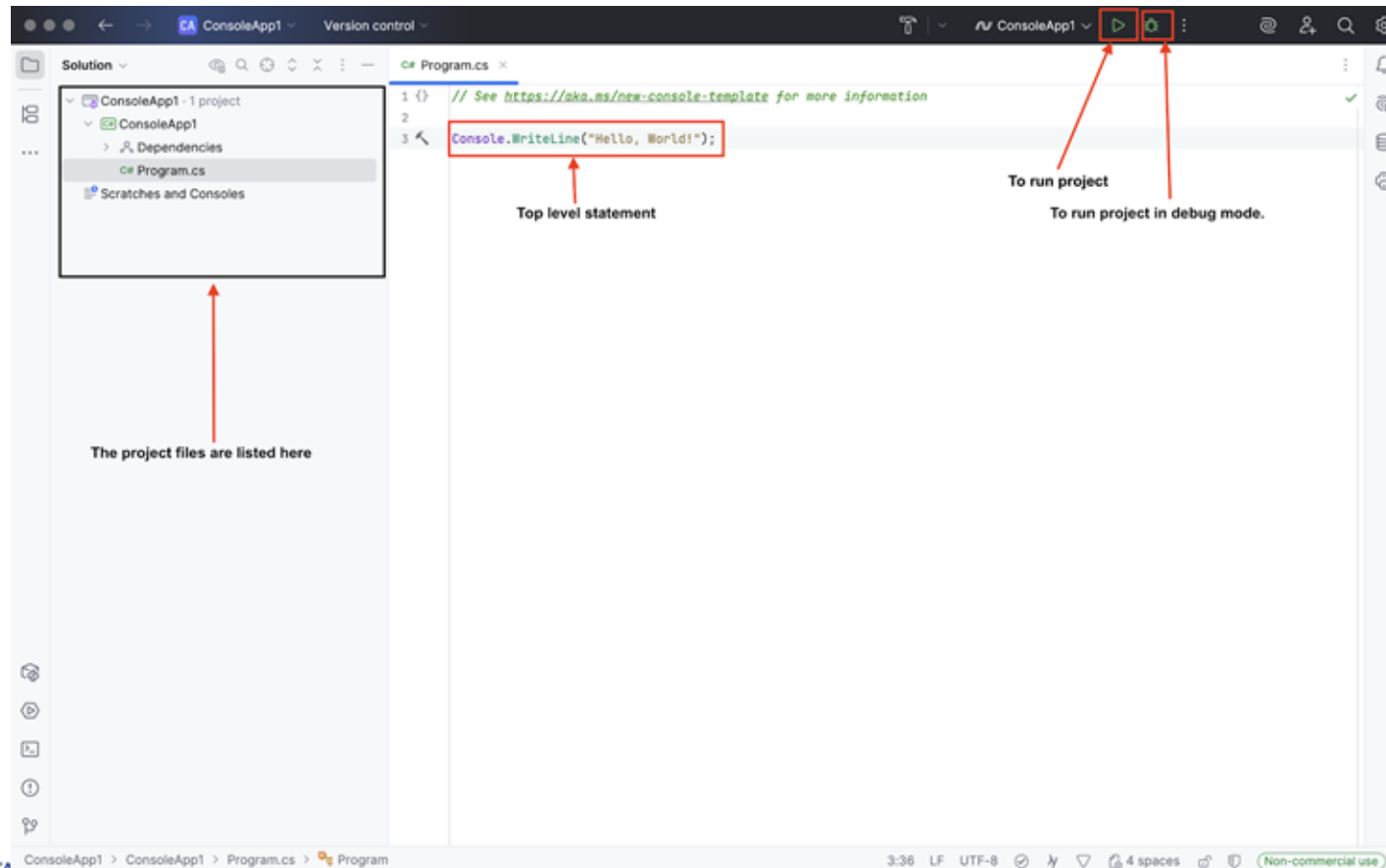
- After that select the console app, choose .NET 9 and select the directory before confirming.



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# Creating a Console App (Rider)

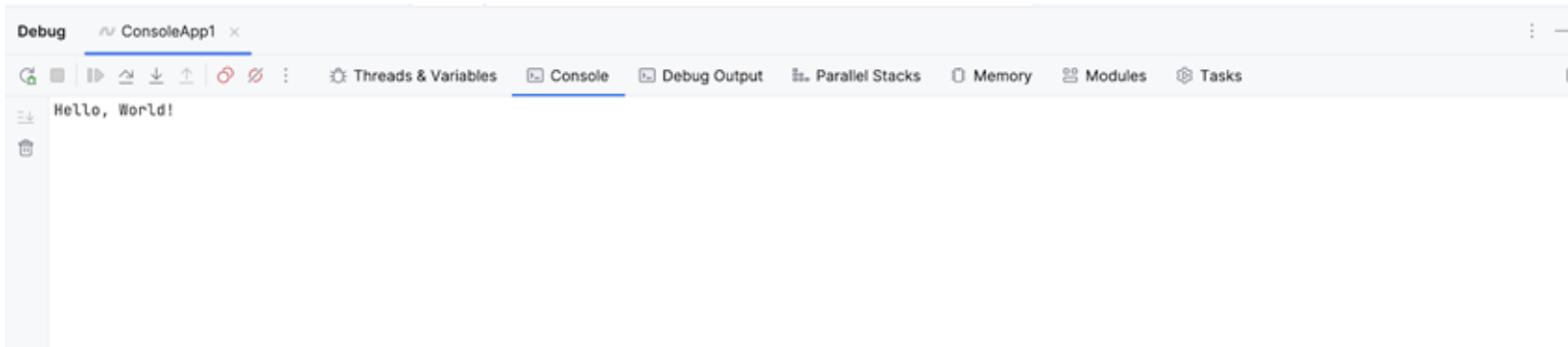
- We can observe that the project has been created and for other descriptions the image attached can be taken for reference. After exploring, hit the debug button as mentioned in the image.



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# Creating a Console App (Rider)

- It can be observed that the hello word was logged in console.

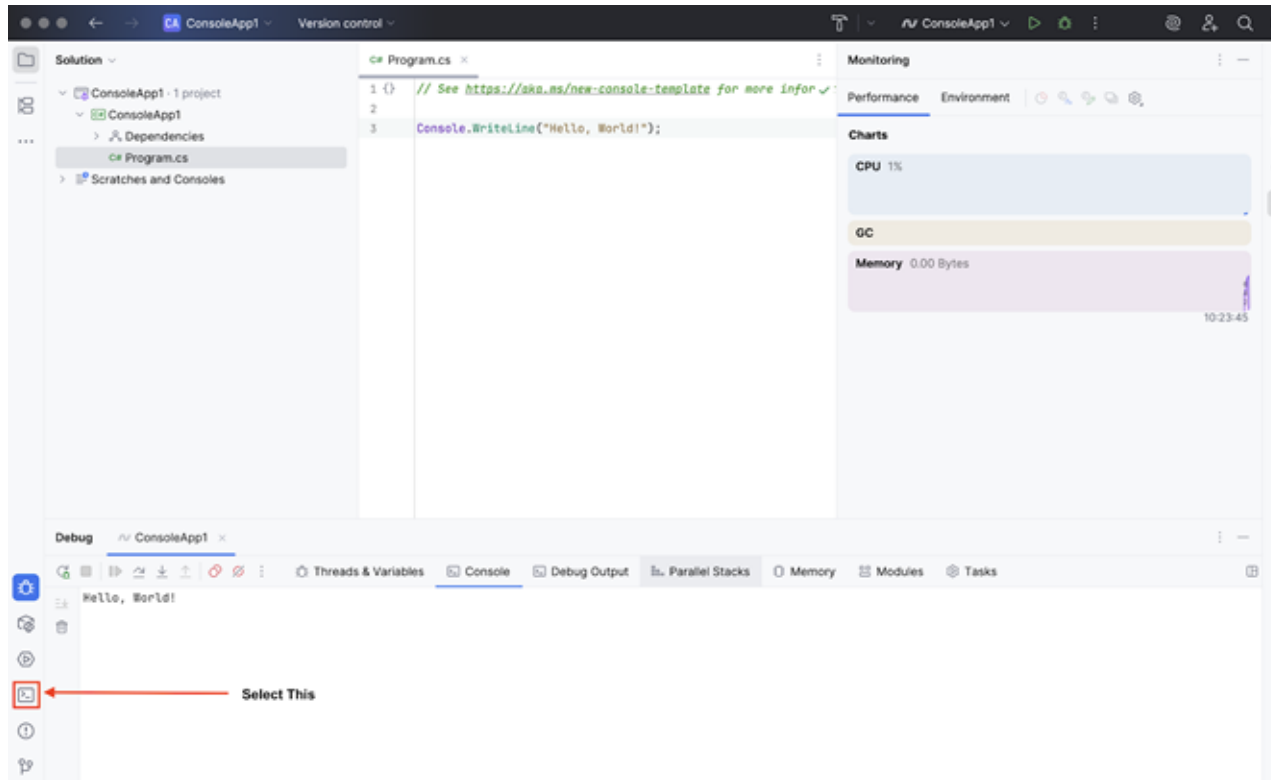


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# Creating a Console App (Rider)

## Running Console App from CLI (Rider)

Navigate to the terminal on the bottom left side of the IDE.



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# Creating a Console App (Rider)

## Navigating to project's directory

Navigate to the terminal on the bottom left side of the IDE. Take reference from image attached below.

- Run **ls** to check if you're in the project directory.
- If not run the command **cd** *project\_folder\_name*

```
aaryanjha@Aaryans-MacBook-Air ConsoleApp1 % ls
ConsoleApp1      ConsoleApp1.sln

aaryanjha@Aaryans-MacBook-Air ConsoleApp1 % cd ConsoleApp1

aaryanjha@Aaryans-MacBook-Air ConsoleApp1 % ls
ConsoleApp1.csproj  Program.cs          bin                 obj
```

**Note:** When Inside the project folder, run ls again to view its contents.

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# Creating a Console App (Rider)

## Running through CLI

Run the command **dotnet run** to run the project.

```
aaryanjha@Aaryans-MacBook-Air ConsoleApp1 % dotnet run  
Hello, World!
```

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# Lab Task

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1. Install **Visual Studio 2022** or **Rider** in your computer System. [ **40 Minutes**]
2. Create a **Simple Printing Console Application** using .NET 8/9.
  - The application should simply **print** your personal details (like **name, address, contact number, college, etc.**) on the console. [10 Mins]
3. Additional Task:
  - Create another class that stores and prints a **motivational quote of the day**.
  - Use **variables** to store the quote before printing it. [10 Mins]

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# Any Questions?